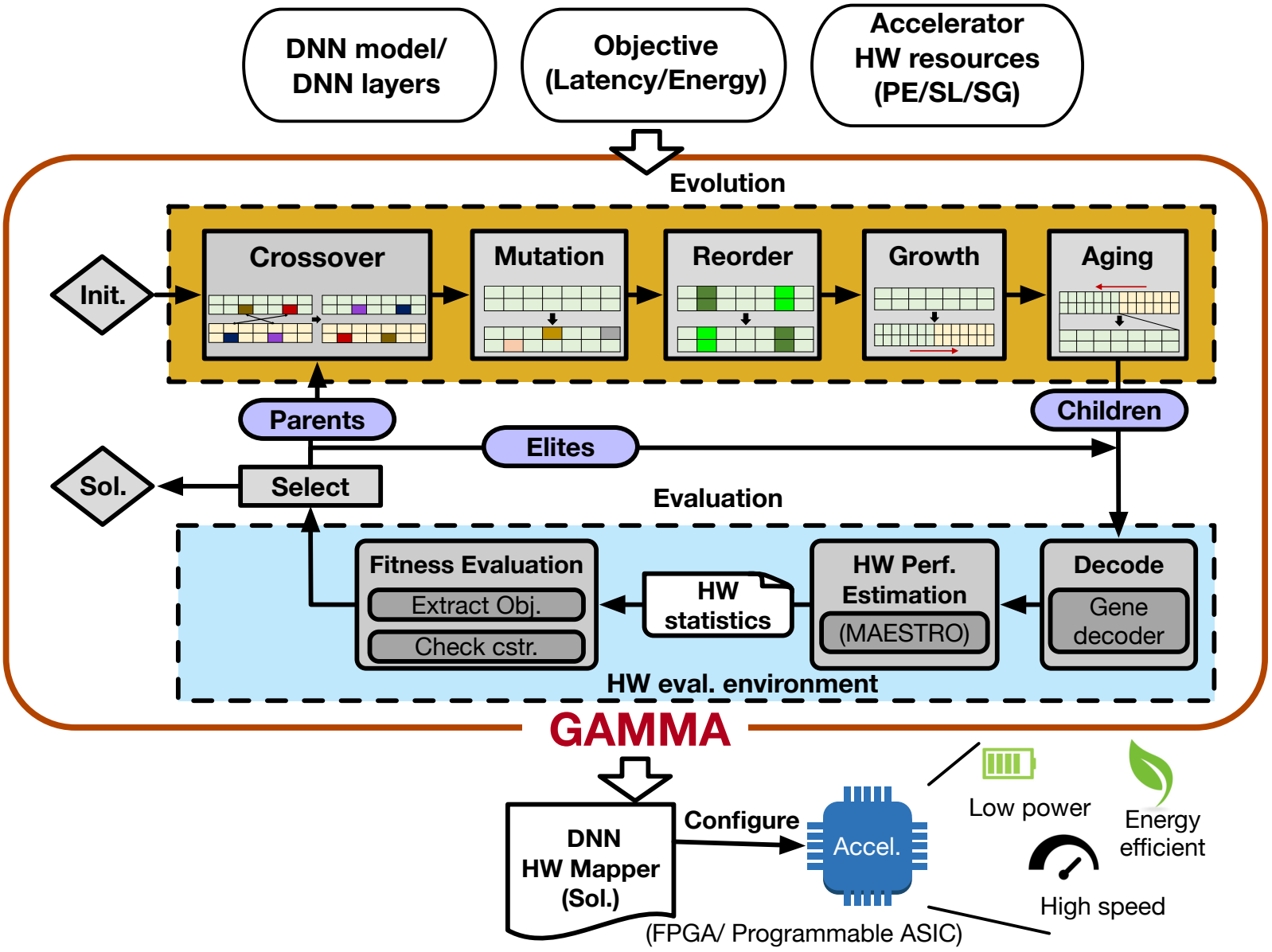


(a) Structure and algorithm flow

| Optimization methods | | Description | Mapping Space Aspects | | | |
|---|---|---|---|---|---|---|
| | | | **Tile Strat.** | **Com. Ord.** | **Par. Dim.** | **Par. Level** |
| **Baseline Methods** | | Defined a ***Par. Level***. Encode ***Tile Strat., Com. Ord., Par. Dim.*** into fixed length inputs, and evolve them with algorithms. | ✓ | ✓ | ✓ | ✗ |
| **GAMMA** | Crossover | Randomly pick two genomes from the parent subset. Interchange the value of their tile size. | ✓ | | | |
| | Mutation | Mutate Dim.: Sample a new Par. Dim. Mutate Tile size: Randomly pick a gene pair, sample a new tile size. | ✓ | | ✓ | |
| | Reorder | Randomly pick two gene pairs. Swap their order. | | ✓ | | |
| | Growth | Concatenate a randomly initiated L1-mapper at the tail. | | | | ✓ |
| | Aging | Remove the L1-mapper at the tail. | | | | ✓ |
| | Summary | | ✓ | ✓ | ✓ | ✓ |

(b) The summary of evolution in GAMMA